

Genome-wide Transcriptional Consequences of MBD6 Haploinsufficiency in a Neuronal Model of Autism Spectrum Disorder

Problem Statement

Autism spectrum disorder (ASD) is a neurological developmental condition characterized by impairments in social communication and restricted behaviors. Although ASD is highly heritable, the molecular mechanisms linking genetic perturbations, particularly in chromatin regulation and transcriptional control, to downstream cellular dysfunction remain incompletely understood. Among epigenetic regulators, members of the methyl-CpG binding domain (MBD) protein family, particularly MBD5 and MBD6, are frequently disrupted in individuals with ASD. However, their functional roles in neuronal contexts have been largely unknown or poorly characterized. This gap in our knowledge limits a comprehensive understanding of how these genetic disruptions contribute to the underlying pathology of ASD.

Significance and Background

ASD represents a significant global public health concern, affecting approximately one in every 31 children (Shaw et al., 2025). Although ASD has a strong genetic component, with twin studies indicating concordance rates of 64-91%, its etiology involves a complex interplay between genetic susceptibility and environmental influences (Tick et al., 2016). Notably, the molecular architecture of ASD is highly heterogeneous, involving numerous genes that converge on shared biological pathways rather than a single causative mechanism. This convergence suggests that diverse genetic alterations may disrupt common regulatory systems essential for brain development. (Lombardo et al., 2019).

Among these systems, chromatin regulation and transcriptional control have emerged as key unifying mechanisms in ASD pathogenesis. Increasing evidence from genomic and epigenomic studies indicates that mutations in chromatin regulators, histone-modifying enzymes, and non-coding RNA pathways, as well as disruptions in higher-order chromatin organization, contribute to disease risk (He et al., 2025; Yan, 2024). Despite the identification of many ASD-associated chromatin regulators, the molecular mechanisms linking these genetic perturbations to downstream cellular dysfunction remain incompletely defined (Cukier et al., 2010; Williams et al., 2010).

The identification of MECP2, encoding methyl-CpG-binding protein 2, as the causal gene for Rett syndrome in 1999 provided the first compelling evidence that epigenetic regulation plays a central role in autism-related neurodevelopmental disorders (Amir et al., 1999). Since then, additional members of the MBD protein family have been implicated in ASD susceptibility, highlighting the broader importance of methylation-associated transcriptional regulation in neurodevelopment (Nagarajan et al., 2006).

Unlike MECP2, MBD5 and MBD6 lack key amino acids required for CpG recognition and are therefore considered incapable of directly binding methylated CpG DNA (Laget et al., 2010). Instead, both proteins contain a central proline-rich intrinsically disordered region (IDR), enabling them to function as transcriptional scaffolds through interactions with multiple transcription factors and chromatin-associated proteins. Consistent with this role, recent proteomic studies have identified MBD5 and MBD6 as components of the Polycomb repressive deubiquitinase (PR-DUB) complex, which includes BAP1 and ASXL1/ASXL2 (Tsuboyama et al., 2022). Furthermore, large-scale whole-genome sequencing studies in ASD patients have

identified MBD5 and MBD6 among 134 autism-associated genes, often with frameshift and premature stop-codon mutations indicative of haploinsufficiency (Trost et al., 2022).

Consistent with their proposed regulatory roles, MBD5 and MBD6 have emerged as dosage-sensitive regulators whose altered expression is associated with neurodevelopmental abnormalities. Haploinsufficiency of MBD5 is linked to a well-defined syndrome characterized by intellectual disability, seizures, and autistic features, with copy number variations at the 2q23.1 locus establishing MBD5 causal gene in ASD and epilepsy (Mullegama et al., 2014; Talkowski et al., 2011). Disruption of MBD5 is also associated with a broad spectrum of neuropsychiatric phenotypes, highlighting the pleiotropic effects of chromatin-associated regulators on brain function and behavior (Bonnet et al., 2013; Hodge et al., 2014). In contrast, although MBD6 shares structural and functional similarities with MBD5, its role in ASD remains comparatively underexplored (Cukier et al., 2010).

In parallel, emerging evidence links mitochondrial dysfunction to ASD pathogenesis. Studies have reported altered mitochondrial morphology, reduced oxidative phosphorylation capacity, and impaired ATP production in brain tissue and other cell types from individuals with ASD. The prevalence of mitochondrial disease in ASD cohorts is estimated to be significantly higher than the general population. Key mitochondrial genes including COX17 (cytochrome c oxidase assembly factor) and PDK1 (pyruvate dehydrogenase kinase 1) have been implicated in regulating the balance between oxidative phosphorylation and glycolysis, which is a balance known to be critical during neuronal development and function (Shulyakova et al., 2017).

Despite these advances, the cellular and transcriptional consequences of reduced MBD6 gene dosage in neuronal contexts, and their potential role in linking epigenetic dysregulation to mitochondrial dysfunction, remain poorly understood. Elucidating these mechanisms is essential for understanding how disruption in chromatin-associated regulators contribute to ASD pathogenesis at the cellular level.

Aims and Rationale

The first aim of this study was to determine the transcriptional consequences of MBD6 haploinsufficiency in human neuronal cells. MBD6 is a chromatin-associated transcriptional regulator, yet its downstream gene targets and regulatory effects remain poorly defined in neuronal contexts. Utilizing microarray-based transcriptomic data from MBD6-heterozygous SH-SY5Y cells, which model the heterozygous loss-of-function observed in ASD patients, enables systematic identification of genes whose expression is altered due to reduced MBD6 dosage. It is hypothesized that MBD6 haploinsufficiency will result in widespread transcriptional dysregulation, characterized predominantly by gene upregulation due to loss of chromatin-mediated repression.

The second aim of this study was to identify the biological pathways and functional systems disrupted by MBD6 haploinsufficiency, especially since differential expression alone doesn't capture higher-order biological processes affected by gene dysregulation. Functional enrichment analyses provide a systems-level interpretation of transcriptional changes by identifying overrepresented biological processes, molecular functions, and signaling pathways among differentially expressed genes (DEGs). It is hypothesized that MBD6 haploinsufficiency will lead to significant enrichment of genes involved in mitochondrial function oxidative phosphorylation, energy metabolism pathways, and neurodevelopmental signaling pathways.

The third aim of this study was to evaluate the genetic and disease relevance of MBD6-associated transcriptional changes. Linking gene expression changes observed in a cellular model

to population-level genetic variation strengthens their relevance to ASD pathogenesis. By integrating DEGs with expression quantitative loci (eQTL) data from GTEx brain and blood tissues, this analysis assesses whether genes dysregulated by MBD6 haploinsufficiency are under genetic control in human populations and whether these regulatory variants overlap with ASD-associated loci identified through genome-wide association studies (GWAS). This approach bridges molecular and genetic level of analysis, moving beyond cell-based observations to broader disease mechanisms. It is hypothesized that eQTLs associated with MBD6-dependent DEGs will significantly overlap with ASD-associated GWAS variants, supporting a convergent genetic and transcriptional basis for ASD risk.

Methods

Publicly available microarray data from the Gene Expression Omnibus (GEO; accession GSE314093) were analyzed, comprising transcriptomic profiles of 2 MBD6-heterozygous and 2 matched control SH-SY5Y human neuroblastoma cells generated using the Agilent 4x44K platform (GPL13497) (Meguro-Horike et al., 2026). SH-SY5Y cells are widely used in vitro neuronal models, making them suitable for investigating transcriptional changes relevant to neurodevelopmental disorders (Devyatov et al., 2025; Lopez-Suarez et al., 2022). The pipeline included data preprocessing and quality control, exploratory data analysis (EDA), differential expression (DE) analysis, gene ontology (GO) enrichment analysis, as well as expression quantitative trait loci (eQTL) integration and GWAS overlap analysis.

To obtain reliable expression estimates, raw Agilent microarray signals were first corrected for background noise using the *normexp* method, which adjusts low-intensity values while preserving true signal. Quantile normalization was then applied across all samples to enforce a consistent distribution of expression values, thereby minimizing technical variability and enabling direct comparison between conditions. Probes were filtered to retain only those with detectable expression above background (*gIsWellAboveBG*) in at least two samples, reducing noise from uninformative features. Platform-specific annotation (GPL13497) was incorporated to map probes to gene symbols, and probes lacking valid annotations were excluded to ensure interpretability in downstream analyses.

Data quality and normalization effectiveness were evaluated through multiple complementary approaches. Distributions of log₂-transformed expression values were examined before and after normalization to confirm consistency across samples. Principal component analysis (PCA) was used to assess global variation in the dataset and to determine whether samples segregated according to experimental condition. Sample-to-sample relationships were further evaluated using Pearson correlation coefficients, visualized as a heatmap to identify clustering patterns and potential outliers. Additionally, MA plots were used to assess intensity-dependent biases and confirm that normalization effectively stabilized variance across expression levels.

To identify genes affected by MBD6 haploinsufficiency, differential expression analysis was performed using a linear modeling framework with empirical Bayes moderation to improve variance estimation (Ritchie et al., 2015). Statistical significance was determined using Benjamini-Hochberg-adjusted p-values, and genes were defined as differentially expressed based on false discover rate or FDR less than 0.05 and absolute log₂ fold change greater than 1 (Ritchie et al., 2015). Redundant probes were consolidated by selecting the most significant probe per gene. Results were visualized using volcano plots and heatmaps with Z-score scaling and hierarchical clustering.

To interpret the biological significance of transcriptional changes, functional enrichment analysis was conducted using GO. Gene identifiers were first standardized by converting gene symbols to Entrez IDs and updating outdated annotations to maximize coverage. GO enrichment analysis was performed for biological process, molecular function, and cellular component categories (Muley, 2025). Statistical significance was determined using Benjamini-Hochberg correction to control for multiple testing. Enriched terms and pathways were visualized using bar plots and network-based representations to highlight relationships among genes and functional categories.

To evaluate the genetic and disease relevance of MBD6-associated transcriptional changes, DEGs were integrated with eQTL data from GTEx v11 brain cortex tissue and ASD-associated variants from the GWAS catalog. Genes exhibiting significant eQTL signals (eGenes; $q \leq 0.05$) were first identified and intersected with the set of DEGs to determine which transcriptional changes may be under genetic regulation (GTEx Consortium, 2020). For these overlapping genes, all associated significant variant-gene pairs were retrieved. To assess their relevance to ASD, genomic coordinates of eQTL variants were compared with ASD-associated loci (EFO_0003756) obtained from the GWAS Catalog, restricting analysis to entries explicitly annotated as autism spectrum disorder (Buniello et al., 2019). Overlap was defined based on physical proximity, with eQTL variants considered associated if located within a ± 500 kb window of ASD GWAS loci, capturing potential linkage relationships. The integrative approach enables identification of regulatory variants that may connect MBD6-dependent transcriptional dysregulation with ASD genetic risk.

Results

Our investigation into the transcriptional and cellular consequences of MBD6 haploinsufficiency in SH-SY5Y human neuroblastoma cells yielded several key findings, some of which supported our initial hypotheses, while others led to a refinement of our understanding.

For data preprocessing and quality control, the pipeline performed as expected. Initial evaluation of raw log₂-transformed intensity distributions indicated high overall data quality across all four samples, in which expression distributions were already highly similar, with median values centered around 7 and comparable interquartile ranges across samples, suggesting minimal technical variability (Fig. 1a). Following quantile normalization, expression distributions across all samples became nearly identical, with complete alignment of medians, interquartile ranges and overall spread (Fig. 1b). This confirms that normalization was effective in removing any residual technical variation while preserving biological signal. Because low-intensity outliers remained present after normalization, probes were then filtered based on signal detection using the gIsWellAboveBG metric, which identifies probes with signal distinguishable from background noise. Retaining probes detected in at least two samples ensured that only reliably expressed features were included. This filtering step reduced the dataset from 44,495 to 30,691 probes, removing approximately 31% of low-confidence probes and thereby enhancing the robustness of subsequent analyses.

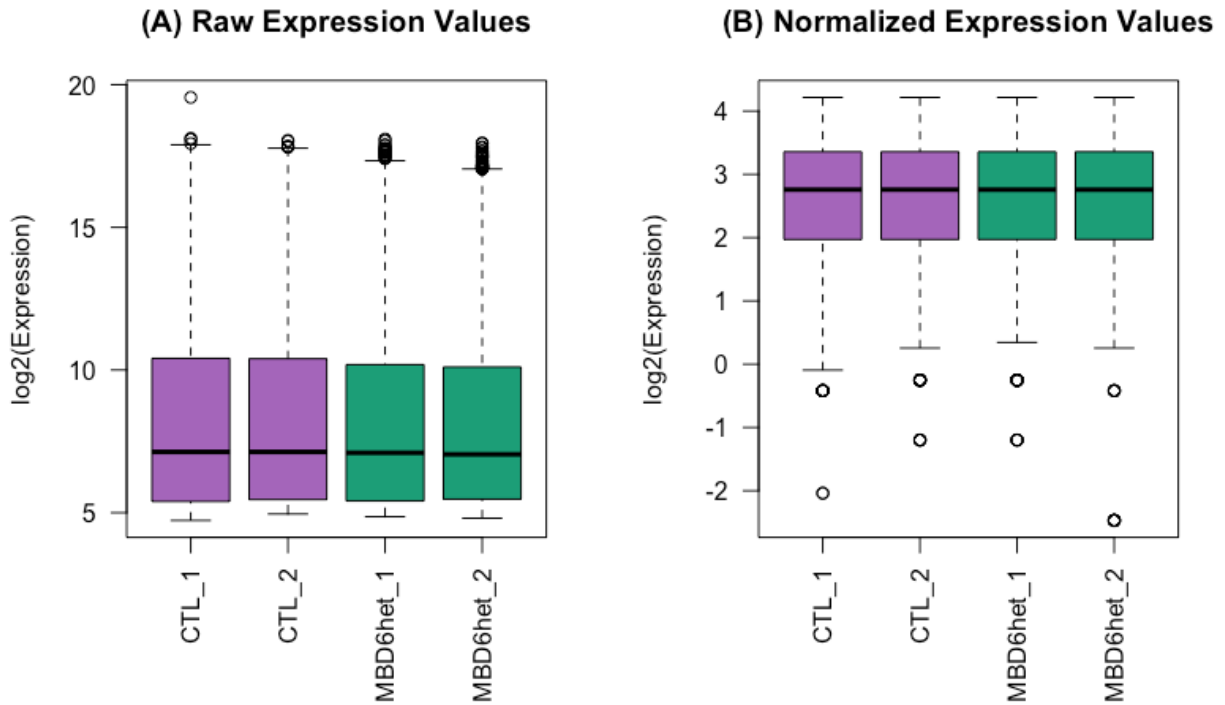


Figure 1. Distribution of Microarray Expression Values Before and After Normalization. (A) Raw log₂-transformed expression intensities across all four samples prior to normalization, showing slight differences in median expression levels between Control (purple) and MBD6-heterozygous (green) samples. (B) Log₂-transformed expression intensities following background correction and quantile normalization, demonstrating successful alignment of expression distributions across all samples.

Having confirmed normalization quality and filtered the dataset to 30,691 high-confidence probes, EDA (PCA, sample correlation analysis, and intensity-dependent bias) was conducted to assess global sample relationships and identify any sources of systematic variation. PCA separated MBD6-heterozygous and control samples along PC1 (principal component 1), which explained 66.9% of total variance, confirming that MBD6 haploinsufficiency is the dominant source of transcriptional variation (Fig. 2). Although the two MBD6-heterozygous replicates clustered tightly together, demonstrating high reproducibility, control replicates showed greater separation along the second principal component (PC2, 22.5% variance) (Fig. 2). This unexpected observation suggests minor within-group variability; thus, given the limited number of replicates, this variability is noted as a potential limitation.

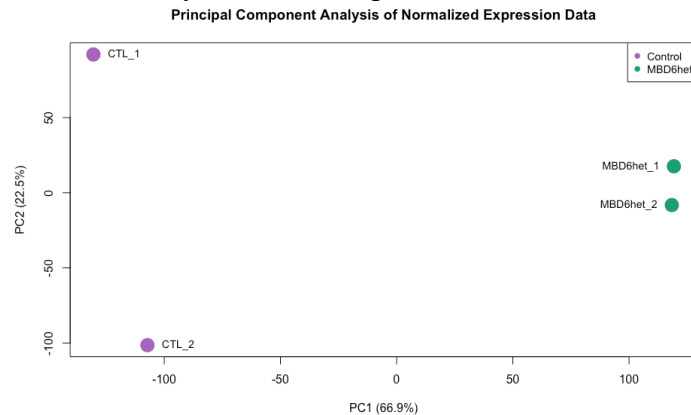


Figure 2. Principal Component Analysis (PCA) of Normalized Microarray Expression Data. Each point represents one sample, colored by condition.

Sample correlation analysis further supported the overall consistency of the dataset. Pearson correlation coefficients were high within each condition, with values of 0.996 for control samples and 0.997 for MBD6-heterozygous samples, indicating strong reproducibility (Fig. 3). Between-group correlations were slightly lower (0.979-0.981), consistent with the presence of biologically meaningful differences between conditions (Fig. 3). No outlier samples were identified, supporting the suitability of the dataset for downstream analysis.

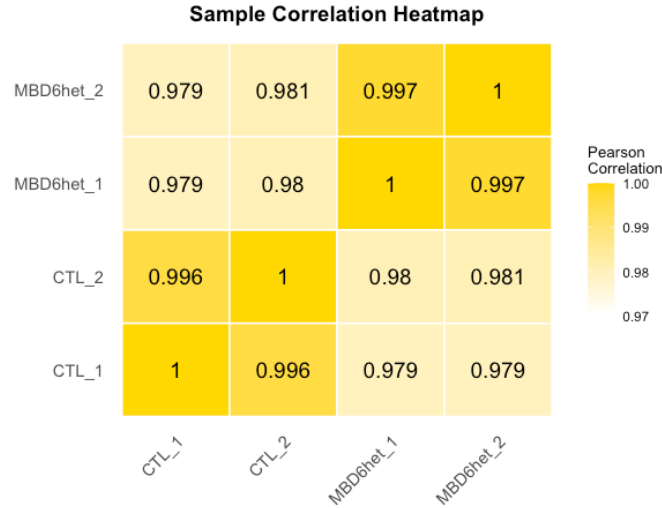


Figure 3. Pearson correlation heatmap of normalized microarray expression values across all four samples. Color intensity represents the Pearson correlation coefficient, with darker yellow indicating higher correlation.

MA plots were used to evaluate intensity-dependent bias following normalization. Across all samples, probes were symmetrically distributed around $M = 0$, indicating that normalization successfully corrected for systematic bias (Fig. 4). A characteristic funnel-shaped distribution was observed, with increased variability at lower intensity values, which is typical for microarray data. Among the samples, CTL_1 exhibited a slightly broader spread and a small number of probes with larger deviations from zero, consistent with its separation along PC2 in the PCA analysis. However, no systematic trends or curvature were observed, and all samples were retained for further analysis (Fig. 4).

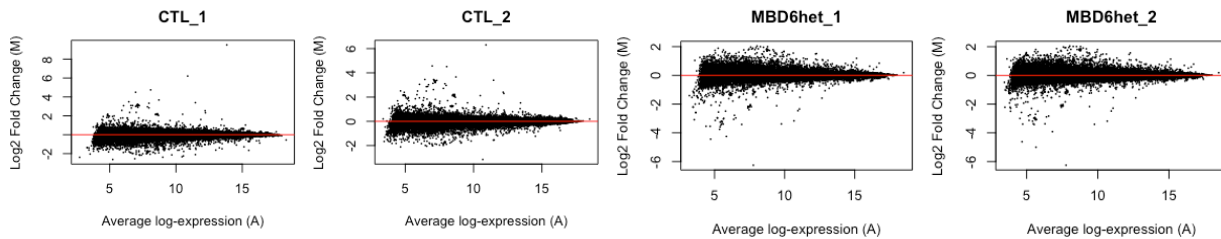


Figure 4. MA plots of normalized microarray expression values for all four samples. Each point represents one probe, with the x-axis showing the average log₂ expression intensity (A) and the y-axis showing the log₂ fold change relative to the array mean (M). The red horizontal line at $M = 0$ indicates no deviation from the array mean.

With the dataset confirmed to be of high quality and samples clearly separating by condition, differential expression analysis was performed to identify specific genes whose expression was altered by MBD6 haploinsufficiency. A total of 913 DEGs were identified at an FDR-adjusted p-value < 0.05 and an absolute log₂ fold change ($|\log_2FC| > 1$) (Fig. 5). Among these, 637 genes were upregulated and 276 were downregulated in MBD6-heterozygous samples compared to controls, resulting in a 2.3:1 ratio of upregulated to downregulated genes (Fig. 6). This predominant upregulation of genes was consistent with the hypothesis that MBD6 functions

primarily as a transcriptional repressor (Meguro-Horike et al., 2026). Top downregulated genes included neuronal function genes (GAL, NPY, MYO16, UNC5D) that supports the neuronal gene dysregulation component of the hypothesis (Fig. 6). However, mitochondrial genes were not prominently represented among the statistically significant DEGs, which was unexpected given the proposed hypothesis of mitochondrial dysfunction as a central mechanism.

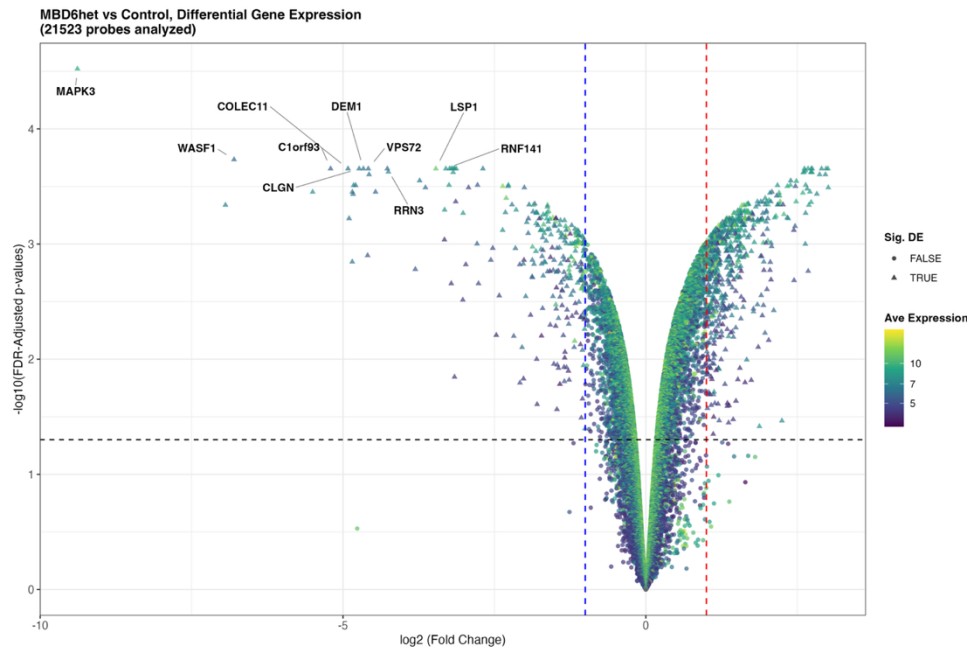


Figure 5. Volcano plot of DEGs between MBD6-heterozygous and Control samples across 21,523 probes. The x-axis represents the log2 fold change (MBD6het relative to Control) and the y-axis represents the -log10 FDR-adjusted p-value. Point shape indicates significance status, with triangles representing non-significant probes. Point color reflects average expression intensity across all samples, with yellow indicating higher expression and dark purple indicating lower expression. Vertical dashed lines mark the log2 fold change thresholds of -1 (blue) and +1 red), and the horizontal dashed line marks the FDR significance threshold of 0.05. The top 10 most significant DEGs are labeled by gene symbol.

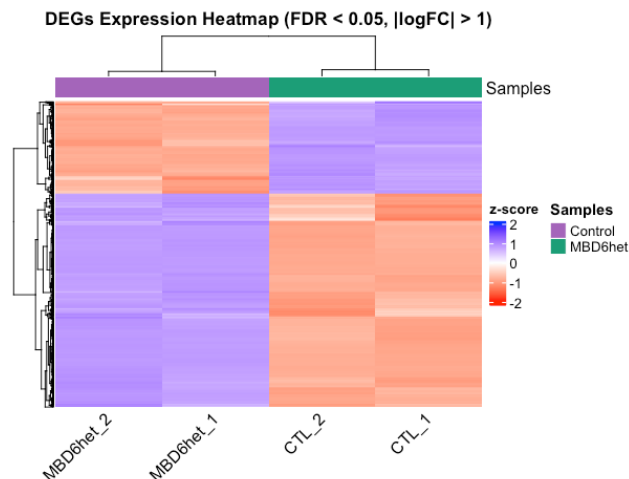


Figure 6. Gene expression heatmap of DEGs. Expression values are row-scaled as Z-scores, where red indicates relatively higher expression and blue indicates relatively lower expression compared to each gene's mean across all samples. Rows represent individual DEGs and columns represent samples, with both rows and columns clustered using hierarchical clustering. The annotation bar at the top indicates sample condition, with purple representing Control replicates (CTL_1 and CTL_2) and green representing MBD6-heterozygous replicates (MBD6het_1 and MBD6het_2).

To move beyond individual gene-level observations and interpret the broader biological significance of these 913 DEGs, functional enrichment analysis using GO was performed to identify overrepresented biological processes, molecular functions, and cellular components. GO enrichment analysis diverged from expectations, as no terms reached the standard FDR < 0.05 threshold. At a relaxed threshold (FDR < 0.10), 28 GO Biological Process terms were enriched across 866 mapped DEGs, clustering into four major biological themes: immune cell activation and differentiation, ERK/MAPK signaling, developmental maturation, and stress-activated kinase signaling (Fig. 7a). Barplot results highlighted regulation of cell activation and leukocyte differentiation (highest GeneRatio), alongside enrichment of ERK/MAPK and NF- κ B signaling pathways, which are implicated in neurodevelopment and ASD (Wen et al., 2014).

Furthermore, gene-concept network analysis showed that immune activation signals were driven primarily by upregulated genes (e.g., TGFB1, BCLL2, BRAF, HLA-DRB1), indicating chromatin-mediated derepression of immune pathways (Fig. 7b). MAPK3, the most significant DEG, acted as a central hub linking MAPK-related pathways, while BRAF bridged immune and MAPK clusters, suggesting convergence of dysregulated processes (Fig. 7b). The absence of mitochondrial pathway enrichment likely reflects the limited statistical power of two biological replicates per condition and the broad chromatin regulatory role of MBD6 across diverse gene sets rather than convergence on a single pathway (Schurch et al., 2016). The immune gene enrichment in a neuroblastoma cell line was unexpected and may reflect shared chromatin accessibility between neuronal and immune gene loci regulated by MBD6, consistent with emerging evidence of neuroimmune dysregulation in ASD (Meltzer & Van de Water, 2017).

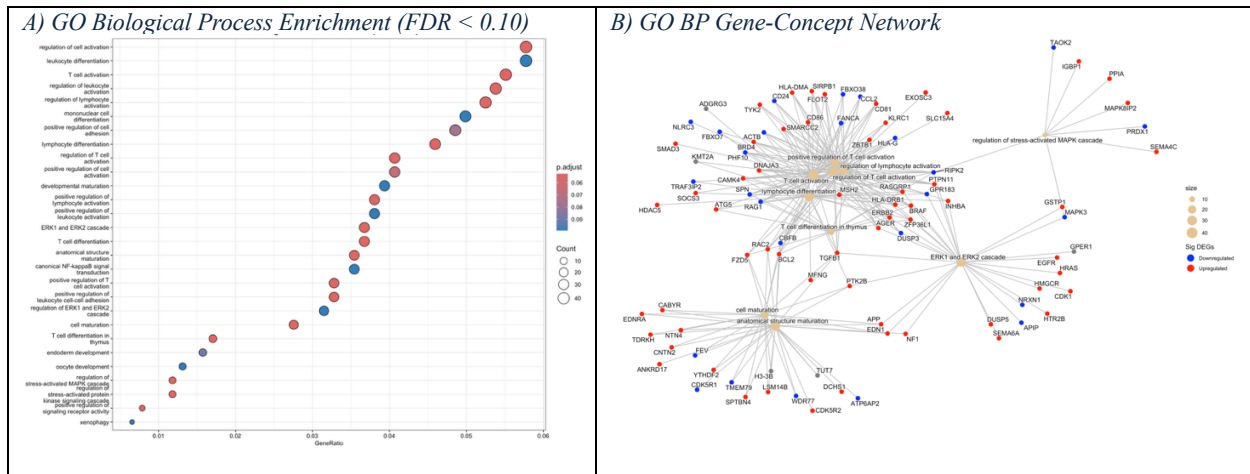


Figure 7. Visualizations for Functional Enrichment Analysis using GO. (A) Dotplot of enriched GO BP terms among significant DEGs from MBD6-heterozygous samples. The x-axis represents the GeneRatio, defined as the proportion of mapped DEGs annotated to each GO term relative to the total number of mapped DEGs (866 genes). Dot size represents the number of DEGs annotated to each term and dot color represents the FDR-adjusted p-value, with red indicating more significant enrichment and blue indicating less significant enrichment. Terms are ranked by GeneRatio in descending order. (B) Gene-concept network of the top 10 enriched GO BP terms among significant DEGs from MBD6-heterozygous samples. Large gold nodes represent enriched GO terms, sized proportionally to the number of DEGs annotated to each term. Small colored nodes represent individual DEGs, with red indicating upregulated genes and blue indicating downregulated genes in MBD6-heterozygous samples relative to controls. Edges connect individual genes to their associated GO terms.

Given that GO enrichment analysis failed to confirm the original mitochondrial dysfunction hypothesis, the study hypothesis was revised to propose that MBD6 haploinsufficiency broadly dysregulates chromatin-mediated gene programs, encompassing both immune activation and mitochondrial pathways, and that eQTL-GWAS integration would

provide a more direct genetic test of this expanded hypothesis. This analytical deviation was justified by the premise that eQTL-GWAS integration provides a stronger and more biologically meaningful connection between transcriptomic findings and ASD genetic architecture than pathway enrichment alone (Porcu et al., 2019). Using GTEx v11 Brain Cortex eQTL data (GTEx Consortium, 2020), 357 of 913 DEGs (39.1%) were identified as significant brain cortex eGenes, and of these, 22 DEG eGenes (6.2%) had eQTL variants within 500kb of ASD GWAS variants from the GWAS Catalog (EFO_0003756), representing 35 unique DEG-GWAS variant pairs. The lollipop plot showed an asymmetric pattern, with 13 upregulated and 9 downregulated genes, consistent with the overall DEG ratio and the repressor function of MBD6 (Fig. 8).

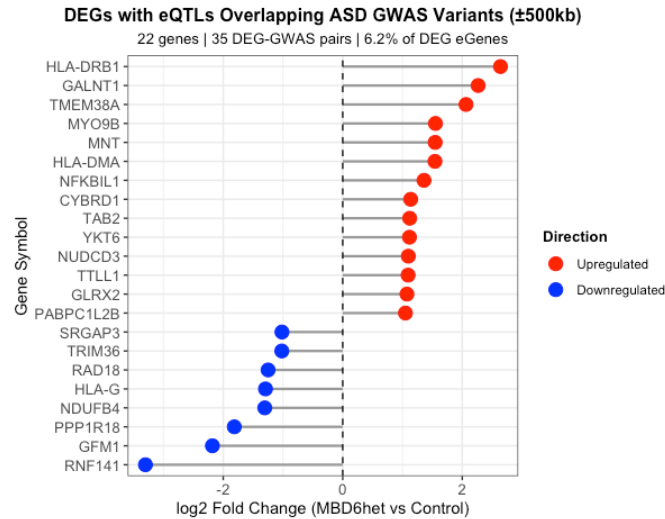


Figure 8. Lollipop plot of log2 fold changes for the 22 DEGs whose eQTL variants overlap with ASD. Each point represents one DEG eGene, positioned on the x-axis according to its log2 fold change in MBD6-heterozygous samples relative to controls. Red points indicate upregulated genes, and blue points indicate downregulated genes in MBD6-heterozygous samples. Horizontal lines extend from zero to the log2 fold change value for each gene, with the vertical dashed line at $x = 0$ serving as reference. Genes are ranked from most downregulated (bottom) to most upregulated (top).

Notably, mitochondrial genes NDUFB4 and GFM1 were downregulated and overlapped with ASD-associated variants, providing direct mitochondrial evidence not captured by GO analysis and supporting the hypothesis that MBD6 haploinsufficiency impairs mitochondrial gene expression (Anitha et al., 2012). The ASD candidate gene SRGAP3 was also downregulated and overlapped with GWAS loci, linking transcriptional changes to established ASD risk. SRGAP3, a known ASD candidate gene involved in dendritic spine morphology and synaptic development, was also downregulated and overlapped with ASD GWAS variants, further connecting the transcriptional dysregulation observed under MBD6 haploinsufficiency to established ASD genetic risk loci (Carlson et al., 2011). Among upregulated genes, HLA-DRB1 showed the strongest fold change (+2.64 log2FC) and was among the most genetically connected to ASD GWAS variants, reinforcing the immune activation signal identified in GO enrichment analysis and suggesting that chromatin derepression of MHC class II immune genes under MBD6 haploinsufficiency is independently supported by ASD genetic architecture. Together these findings confirm the revised hypothesis that MBD6 chromatin regulation controls both mitochondrial gene expression and immune activation gene programs, and that both programs are independently connected to ASD genetic architecture through separate sets of eQTL variants overlapping ASD GWAS loci.

In conclusion, this study addressed the poorly understood functional role of MBD6 in neuronal contexts by characterizing the genome-wide transcriptional consequences of MBD6 haploinsufficiency in SH-SY5Y human neuroblastoma cells. Differential expression analysis revealed that reduced MBD6 dosage produces widespread transcriptional dysregulation, predominantly through upregulation of target genes, consistent with MBD6 functioning as a chromatin-mediated transcriptional repressor. Functional enrichment and eQTL-GWAS integration together identified two molecular mechanisms linking MBD6 disruption to downstream cellular dysfunction: derepression of immune activation gene programs, including MHC class II and MAPK signaling pathways, and impaired expression of mitochondrial genes including NDUFB4 and GFM1. Critically, both mechanisms were independently supported by ASD genetic architecture through eQTL variants overlapping ASD GWAS loci, and the downregulation of the established ASD candidate gene SRGAP3 further connected MBD6-dependent transcriptional changes to synaptic dysfunction. Collectively, these findings suggest that MBD6 haploinsufficiency simultaneously disrupts neuroimmune and mitochondrial gene regulatory programs, providing a plausible molecular basis through which reduced MBD6 dosage may contribute to ASD pathogenesis and advancing our understanding of how chromatin regulatory disruption translates into the cellular dysfunction underlying neurodevelopmental disorders.

Limitations and Alternative Approaches

The primary limitation of this project is that the dataset includes a limited number of biological replicates (two biological replicates per condition). Having a small sample size may reduce statistical power and increase sensitivity to technical variability. To mitigate this issue, the analysis will employ empirical Bayes variance moderation to stabilize variance estimates across genes and improve inference reliability. Second, microarray technology has inherent constraints, including limited dynamic range and potential probe cross-hybridization, which may affect detection accuracy for low-abundance transcripts. Pathway-level analyses will help reduce noise by focusing on coordinate biological patterns rather than isolated gene-level effects.

Additionally, SH-SY5Y cells represent an in vitro neuronal model and may not fully recapitulate the complexity of human brain tissue. As an alternative or complementary approach, findings could be validated using independent transcriptomic datasets, RNA sequencing data with greater sensitivity, or targeted experimental validation such as quantitative PCR for key genes of interest. Sensitivity analyses exploring different statistical thresholds will also be performed to ensure robustness of conclusions. These strategies will strengthen confidence in the results and address potential methodological constraints.

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